Executive Summary

Egypt EON Green

Ammonia Project



Hari Ravindran
SVP & Global Head
KBR Technology Solutions

As the executive sponsor, I am excited to support ACWA Power's vision to develop the Egypt EON Green Ammonia Project.

This is a strategic project for both ACWA Power and Egypt, as it positions the country to become a key player in the global green hydrogen economy.

KBR is privileged to collaborate with you and deliver its cutting-edge technology solutions to drive the project towards successful execution. Our commitment to providing high-quality, cost-effective technology solutions makes us trusted technology partners and advisors to clients across the world. We look forward to working with you as your technology partner on this project.

I am fully committed to providing you with our strongest technology team. We will leverage our expertise in ammonia technology, combined with our best-in-class engineering methodologies, to deliver an optimized design that reduces schedule, capex, and opex, improves reliability and availability, and enhances the project's business case and profitability.

Why KBR?

WORLD-LEADING GREEN AMMONIA TECHNOLOGY PROVIDER
Awarded 10+ licensing and basic engineering design contracts

EXPERIENCE IN DELIVERY OF PROJECTS IN EGYPT
Awarded 4 grassroots ammonia projects over the last 10 years

SUPPORTING FID ACHIEVEMENT
Ensuring bankability by utilization of proven design operating in similar capacities

EFFICIENT AND RELIABLE SYNTHESIS LOOP
Lowest high-pressure equipment count, leading to lower capex and opex

PROPRIETARY DIGITALIZATION SOLUTIONS
KBR will deliver advanced process control system, remote monitoring, operating training simulator and other tools to improve LCOA

OPERATIONAL FLEXIBILITY
Rapid ramp down/up of the ammonia unit with load variation of 1.5% per minute and capability of 10% turndown

ONE KBR DELIVERY SOLUTION
Potential for optimized cost and schedule for technology PED & FEED by One KBR

Project Excellence

WORLD LEADER

- In ammonia technology, **260+ grassroots plants since 1944**
- In green ammonia, **10+ awarded projects**

WORLD's MOST

- **Energy-efficient** ammonia plant **6.27Gcal/MT**
- **Reliable** ammonia plant **2,162 consecutive days** in operation

WORLD's LARGEST

Ammonia plant with single converter 3,000 MTPD

Our Projects Track Record

KBR has a proven track record of delivering large and complex hydrogen and green ammonia projects across the world including Egypt.

These projects include complex interfaces, local regulatory requirements, protection of licensor technology.

Our tools, best practices and systems allow us to transfer knowledge, lessons learnt and best practice as well as de-risk potential threats and realize opportunities to your project.



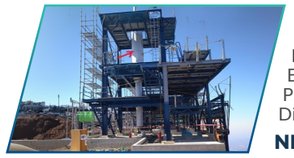
PreFEED
HYPORT DUQM

Development of a Green hydrogen production facility utilising Electrolyser technology in the Special Economic Zone at Duqm, Oman.



Owners Engineering
MADOQUA: POWER2X

KBR will provide technology license, proprietary engineering design and equipment, and catalyst solutions to Madoqua for its green ammonia projects, including 1,200 tonnes per day in Phase 1 and 1,500 tonnes per day in Phase 2.



License, Basic Engineering, Proprietary Equipment, Digital Solutions
NEDO-JGC

Construction on-going for 4 MTPD green ammonia. To be operational in 2025, includes KBR Advance Process Control (APC).



License, Basic Engineering, Proprietary Equipment
KIMA

Utilization of KBR's Purifier ammonia process with low overall energy consumption.



License, Basic Engineering, Detailed Engineering, Procurement and Construction
EBIC

World's largest application of KBR's proprietary ammonia technology, KAAP™ since its introduction to the market in 1992, and the first in Egypt. KBR supported the development, project financing and continuous digital services using KBR Insite tool for monitoring performance and plant reliability.



License, Basic Engineering, Proprietary Equipment
ACME

Basic engineering and proprietary equipment supplied for 300 MTPD ammonia synloop.



Paolo Brunengo
Technology Director

- Technology leader in syngas production and in the blue and green hydrogen and ammonia value chain
- Initiated and led the development of a proprietary technology solution for green ammonia (K-GreeN®)

Collaborative and Experienced Project Team Committed and Available for Egypt EON BEDP Execution and Program Support

A Highly Collaborative & Experienced Team

Project Manager
Mei Catchpole



- Project Manager with extensive experience in mega projects
- Experienced in full lifecycle execution as engineering manager
- Knowledge of process design, plant operations and process safety

Program Director
Andrew Delugar



- Director of complex/mega projects and programs of work
- Project Manager on ammonia and ammonia cracking PEDs
- Engineering Manager with execution based around delivery of standardised building blocks

Technology Lead
Rohit Khurana



- Subject Matter Expert in design and optimisation of ammonia syngas projects
- Process Lead for green ammonia technology development
- Site commissioning and performance test experience

Process Lead
Noelia Zambrano



- Technical Professional Leader with 20+ year experience in design of chemical plants
- Process Lead for execution of proprietary design packages for green ammonia plants, such as FFI Holmaneset
- Significant experience in working with FEED contractors

Machinery Lead
Prachi Priya



- Experience across full range of rotating equipment
- Recent lead roles on green ammonia and blue hydrogen projects
- Experience with machinery items in licensor, client and EPC roles

KBR's Ability to Support Through Project Lifecycle (ONE KBR)

KBR applies a **truly global approach** for technology project execution with global hubs in the USA, the UK and India and regional offices around the world enabling prompt client collaboration, and project management within the client's regions and time zones (KBR EDGE[SM])

KBR has achieved **98% success rate** for on-time delivery of licensing and basic engineering design (LPED)documentation. KBR also has a strong independent quality control (QC) process, which ensures deliverables are consistent and correct the first time

For further project support, KBR's **Integrated Solution** provides best-in-class program management, FEED, PMC and EPCM services with extensive capabilities and experience in complex and remote project environment

KBR also provides **leading-edge digital tools and solutions** enabling the optimization of operations and maintenance of operating assets to increase production, reduce energy consumption, emissions and opex

KBR has **extensive experience in modularization** with more than 50 years of experience in offshore and onshore environments, green and brownfield sites, covering the full spectrum from simple structural pre-assemblies to worldscale piperack and equipment modules

24/7

All day everyday approach to safety

COURAGE TO CARE

The genuine willingness and authority to intervene

TRANSFORMATIONAL LEADERS IN SAFETY

We have delivered over ***50 million safe hours*** in the last 30 years for a leading energy company

KBR has executed more than **260 ammonia plants** on all continents and has a joint wealth of **experience in engineering and project execution** with our EPC partners. The large EPC partner network of KBR opens access to an unmatched number of fabrication yards around the globe, allowing smooth and fast realization of stick-built plant as well as modular solutions for green and blue ammonia.

Major EPC Partners

CC-7 · TOYO ENGINEERING · JGC · WUHUAN · wison · Tecnimont · SAMSUNG · MCDERMOTT · HYUNDAI ENGINEERING CO., LTD · intecsa industrial · Rekind · Petrofac

Summary

Strong Partnership
KBR can connect ACWA to operators and developers across our extensive network of licensees

Leading Ammonia Synthesis Loop Technology
Compact, reliable, highly flexible, energy efficient and low capex technology with horizontal converter and unitized chiller

Standardized Design
Ammonia synthesis loop applicable to blue and green ammonia and with location-agnostic design approach

World Leader in Ammonia Technology
More than 260 grassroots ammonia plants and 50% market share in global ammonia capacity

Highly Collaborative and Experienced Team
Committed and available for standard BEDP execution

ONE KBR
Integrated technology and FEED execution leading to shorter execution schedule

